Project Report: PassGenius (Python Security Suite)

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Abstract

This report presents PassGenius, an interactive Python command-line tool designed to enhance password security awareness. The tool offers two key features: password strength analysis using the zxcvbn-python library and data breach checking via the Have I Been Pwned (HIBP) API. It provides a user-friendly, color-coded interface with actionable feedback, ensuring passwords meet robust security standards. By combining open-source libraries and a privacy-protecting API query method, PassGenius empowers users to create and verify strong passwords, making it a valuable tool for personal cybersecurity.

1 Introduction

Strong passwords are essential for protecting personal information in todays digital world, yet many users rely on weak or reused passwords. PassGenius is a command-line tool developed to address this issue by analyzing password strength and checking for exposure in known data breaches. It provides clear, immediate feedback to help users improve their security practices. This report outlines the tools used, the development process, and the outcomes, demonstrating how PassGenius enhances cybersecurity through a simple and effective interface.

2 Tools Used

The project was built using the following tools and technologies:

- **Python 3**: Core programming language for the applications logic and functionality.
- **zxcvbn-python**: Dropboxs library for intelligent password strength analysis, evaluating patterns and common words.
- **Rich**: Created a visually appealing command-line interface with colors and formatted tables.
- **PyYAML**: Managed application settings through a config.yaml file for easy customization.
- **Requests**: Handled HTTP requests to query the Have I Been Pwned (HIBP) API.
- **Have I Been Pwned (HIBP) API**: Checked passwords against known data breaches securely.
- **Development Environment**: Visual Studio Code, Linux terminal, Git for version control.

3 Steps Involved in Building the Project

PassGenius was developed through the following steps:

1. **Project Setup**: Initialized a project environment and installed required libraries (zxcvbn, rich, pyyaml, requests) using pip.

2. **Configuration**: Created a config.yaml file to store settings, such as minimum password strength, for easy customization.
3. **Password Strength Checker**: Developed a function using zxcvbn to analyze password strength, displaying scores and suggestions via the rich library.
4. **Breach Checker**: Built a function to:
    - Hash passwords using SHA-1.
    - Send only the hash prefix to the HIBP API.
    - Check the returned hash suffixes for matches.
    - Display clear pwned or safe results.
5. **User Interface Design**: Designed an interactive loop to prompt for passwords (hidden for privacy) and display formatted results using rich.
6. **Final Integration**: Combined all functions into a single script with error handling for issues like network failures or missing configuration files.

4 Conclusion

PassGenius successfully delivers a practical and educational tool for password security, combining intelligent strength analysis with secure breach checking. Its user-friendly interface, powered by the rich library, makes it accessible to all users. By leveraging open-source libraries and the HIBP API, the tool provides actionable feedback to improve password practices. Future enhancements could include password generation features or integration with password managers. This project showcases the power of Python in building impactful cybersecurity tools.